Project Proposal

Submodular Optimization and Greedy Approximation for Distributed Influence Maximization

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1 Topic Selection

We choose **Topic 3: Influence Maximization** from the course reference list. Influence maximization asks how to select a limited number of seed nodes in a network so that information, behavior, products, or interventions spread to as many nodes as possible. It is a standard model for viral marketing and social-network analysis, and it also provides a clean algorithmic connection between modern networked applications and classical approximation algorithms.

This project focuses on **submodular optimization and greedy approximation for distributed influence maximization**. In large networks, no single machine or agent may know the entire graph. Instead, each local agent observes part of the graph or estimates local diffusion effects, then communicates with other agents to approximate a globally effective seed set under a fixed budget.

2 Problem Formulation

Let $G = (V, E)$ be a directed or undirected graph. Each edge (u, v) may have a propagation probability p_{uv}. Given a budget k, the goal is to choose a seed set $S \subseteq V$ with $|S| \leq k$ that maximizes the expected number of activated nodes:

$$\max_{S \subseteq V,\ |S| \leq k} \sigma(S),$$

where $\sigma(S)$ is the expected influence spread from S under a diffusion model such as the Independent Cascade model or the Linear Threshold model. The input is the graph, diffusion parameters, and budget k; the output is a seed set of size at most k.

This is a combinatorial optimization problem. Under common diffusion models, $\sigma(S)$ is monotone and submodular, meaning the marginal gain of adding a new node decreases as the current seed set grows:

$$\sigma(A \cup \{v\}) - \sigma(A) \geq \sigma(B \cup \{v\}) - \sigma(B), \quad A \subseteq B,\ v \notin B.$$

Although exact influence maximization is NP-hard, the standard greedy algorithm achieves a $(1 - 1/e)$ approximation guarantee for monotone submodular maximization under a cardinality constraint.

3 Related Work

Kempe, Kleinberg, and Tardos introduced the influence maximization problem for social networks, proved NP-hardness under the Independent Cascade and Linear Threshold models, and showed how submodular maximization yields a $(1 - 1/e)$-approximate greedy algorithm. This will be the main representative paper for reproduction.

Nemhauser, Wolsey, and Fisher proved the classical approximation guarantee for maximizing a nonnegative monotone submodular function subject to a cardinality constraint. Their result provides the theoretical foundation for the greedy analysis in influence maximization.

Leskovec et al. proposed the Cost-Effective Lazy Forward (CELF) acceleration, which uses diminishing marginal gains to avoid many repeated influence-spread computations. We plan to reproduce both standard greedy influence maximization and CELF-style lazy greedy selection.

4 Algorithm and Technical Plan

We plan to implement and compare the following methods:

- **Random baseline:** choose k seed nodes uniformly at random.
- **Degree baseline:** choose the k nodes with highest degree or out-degree.
- **Greedy influence maximization:** repeatedly choose the node with the largest estimated marginal influence gain.
- **CELF lazy greedy:** maintain upper bounds on candidate marginal gains to reduce unnecessary diffusion simulations.
- **Distributed extension:** partition the graph across multiple agents, let each agent estimate local marginal gains, and aggregate the candidates through limited communication or a consensus-style rule.

The main diffusion model will be the Independent Cascade model, with influence spread estimated by Monte Carlo simulation. Planned datasets include synthetic Erdos–Renyi graphs, Barabasi–Albert scale-free graphs, Watts–Strogatz small-world graphs, and one small real network if time permits. We will measure influence spread, runtime, scalability, and sensitivity to the seed budget k and the number of Monte Carlo trials.

5 Project Scope and Expected Goals

The minimum goal is to reproduce the classical behavior of influence maximization algorithms:

- greedy selection should obtain larger influence spread than random and degree-based baselines;
- CELF should preserve nearly the same solution quality as greedy while reducing runtime;
- influence spread should increase with k, while marginal gains decrease because of submodularity;
- the distributed approximation should illustrate the tradeoff between communication cost and solution quality.

Optional extensions include comparing different propagation probabilities, studying how Monte Carlo sample size affects stability, adding communication-limited distributed selection, or connecting the results to distributed optimization with local information and global objectives.

6 Team Information

Member	Responsibility	Expected Contribution
Yixin Chen	Modeling and theory	Problem formulation and related work
Bichi Zhang	Algorithms	Greedy, CELF, and complexity analysis
Yaoqin Ye	Experiments	Implementation, plots, and distributed extension